Universiteit Antwerpen
Faculteit Toegepaste Ingenieurswetenschappen

5 – Infrastructuur & Beveiliging

Lab 4: VLAN trunking & routing

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1 Preparation

Before attending the lab, complete the following preparation. First, read up on the concept of VLANs, but also the why and the how here: Virtual Local Area Networks (VLANs) [1]. Pay close attention to the wording used in the article to describe the VLANs.

Consider how to do inter-VLAN routing and the concept of a Layer 3 switch: Routing Between VLANs & Layer 3 Switches [2]. You do not need to study the Cisco commands, but it can be useful to consider the output of the commands shown in the tutorial.

Finally, consider the article on Inter-Switch Link and IEEE 802.1Q Frame Format from the website of Cisco [3], but only the part about the IEEE 802.1Q frame. Consider the structure of the IEEE802.1Q frame and where it is located in the IP stack and sub-stacks. It's important that you can understand why there is a limit to the VLAN numbers and how the packet is structured.

2 Lab Goals

In this lab, you will learn how VLANs are used to segment Layer 2 networks, how VLAN traffic is transported across trunk links, and how VLAN communication can be analyzed using Wireshark. The goals of the lab today are the following:

- Configure and verify VLANs on OpenWrt switches.
- Explain how VLANs provide logical network separation.
- Configure and validate access ports and trunk ports.
- Explain and analyze IEEE 802.1Q VLAN tagging.
- Verify connectivity within and between VLANs and explain the observed behavior.
- Configure and verify inter-VLAN routing.

> **When your group finishes an exercise, make sure to alert the lecturer before continuing to the next exercise.**

3 Exercise 1: VLAN configuration

Before beginning this lab, prepare your switches as follows:

1. **Disable the firewall** on all devices to allow all configuration steps.
2. **Install required packages** on each switch:

```
opkg update
opkg install ip-full ip-bridge tc-full
```

> **Make sure that after installing the packages, you revert the newly created WAN interface back to LAN, and make sure to mark it back as untagged in the VLAN settings after reactivating it.**

Configure the network as shown in Figure 1. Each group must create 1 IP network (`10.1.x.x`), using a `/16` subnet mask. Turn on the Spanning Tree Protocol (STP) on all switches before making any physical connections to prevent network loops and broadcast storms. Carefully follow the port and cabling layout indicated in the figure, as this will simplify your VLAN configuration later.

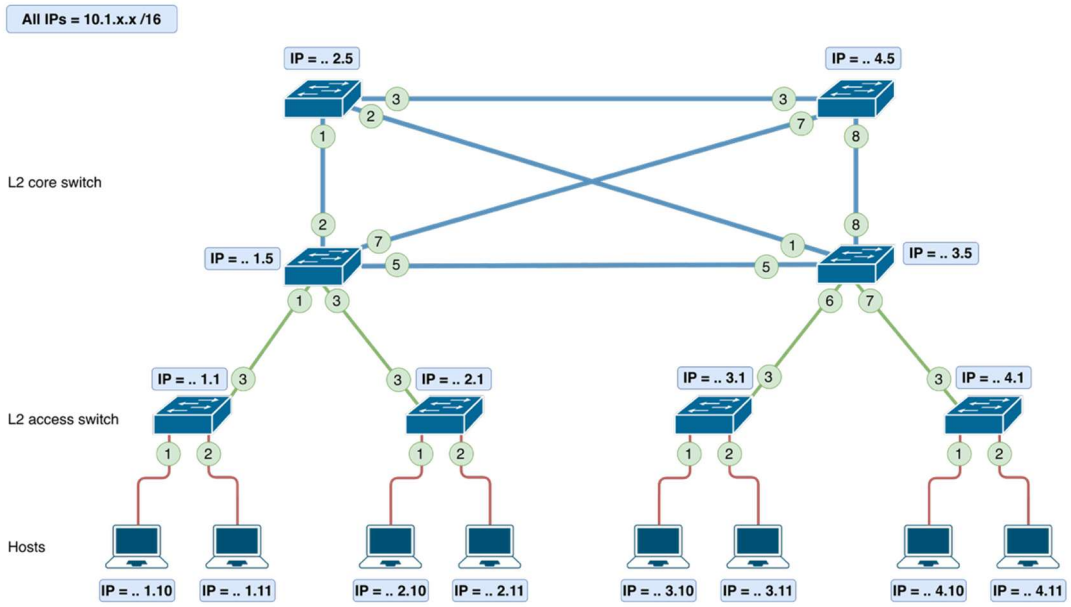


Figure 1: Initial configuration without VLAN separation

> **Make sure all the ports on all the switches are in VLAN1 (= default VLAN) so that you have full connectivity. Only continue if you have full connectivity!**

How will you check this in the GUI, and what about the CLI?

_____

_____

Explain why all the ports should be configured as `untagged`. What does this configuration imply, and what alternative options are available?

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_____

3.1 Configuring VLANs

Now configure VLANs on your switches as shown in Figure 2. When you assign VLANs on your access switches, each port becomes a member of a specific VLAN. **Note that only VLAN1 has the switch's management IP by default** (for example, `10.1.1.1`). This means that only devices connected to ports in VLAN1 can reach the switch's IP, while devices on other VLANs (VLAN2, VLAN3, VLAN4) cannot access the switch unless you specifically assign a management IP address to those VLANs.

> **To manage the switch or make configuration changes, always plug your PC into a VLAN1 port.**

VLAN assignments are indicated by color. You will have to create 4 VLANs per switch:

- **Access switches:** Assign the first five ports to the VLAN corresponding to your switch's subnet. The remaining three ports should be assigned to the other VLANs (one port per VLAN).
    - Access switch `10.1.1.1` will assign:
        - Port 1-5 = VLAN1
        - Port 6 = VLAN2
        - Port 7 = VLAN3
        - Port 8 = VLAN4

Port 1	Port 2	Port 3	Port 4	Port 5	Port 6	Port 7	Port 8
VLAN1	VLAN1	VLAN1	VLAN1	VLAN1	VLAN2	VLAN3	VLAN4

    - Access switch `10.1.2.1` will assign:
        - Port 1-5 = VLAN2
        - Port 6 = VLAN1
        - Port 7 = VLAN3
        - Port 8 = VLAN4

Port 1	Port 2	Port 3	Port 4	Port 5	Port 6	Port 7	Port 8
VLAN2	VLAN2	VLAN2	VLAN2	VLAN2	VLAN1	VLAN3	VLAN4

    - Access switch `10.1.3.1` will assign:
        - Port 1-5 = VLAN3
        - Port 6 = VLAN1
        - Port 7 = VLAN2
        - Port 8 = VLAN4

Port 1	Port 2	Port 3	Port 4	Port 5	Port 6	Port 7	Port 8
VLAN3	VLAN3	VLAN3	VLAN3	VLAN3	VLAN1	VLAN2	VLAN4

    - Access switch `10.1.4.1` will assign:
        - Port 1-5 = VLAN4
        - Port 6 = VLAN1
        - Port 7 = VLAN2
        - Port 8 = VLAN3

Port 1	Port 2	Port 3	Port 4	Port 5	Port 6	Port 7	Port 8
VLAN4	VLAN4	VLAN4	VLAN4	VLAN4	VLAN1	VLAN2	VLAN3

- **Core switches:** Assign two ports per VLAN and follow the figure for the configuration.
    - Port 1-2 = VLAN1
    - Port 3-4 = VLAN2

- Port 5-6 = VLAN3
- Port 7-8 = VLAN4

Port 1	Port 2	Port 3	Port 4	Port 5	Port 6	Port 7	Port 8
VLAN1	VLAN1	VLAN2	VLAN2	VLAN3	VLAN3	VLAN4	VLAN4

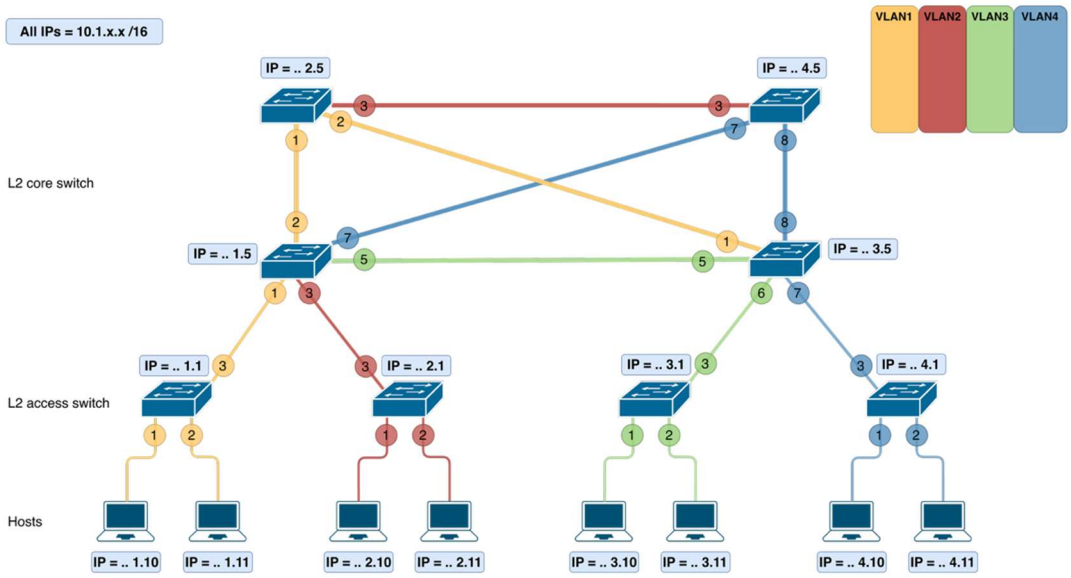


Figure 2: VLAN configuration

How will you do this? Should the ports be tagged or untagged?

_____

_____

Can you still ping everyone on the network?

_____

_____

Pair up with your partner on the access switch. Connect one of your hosts to a different VLAN of the access switch. What is the consequence in terms of reachability?

_____

_____

Pair up with your neighbor on the access switch next to you. Connect the .11 hosts to VLAN4, each on their own access switch. What is the consequence in terms of reachability?

_____

_____

How does a switch make sure VLANs are perfectly separated?

_____

_____

4 Exercise 2: VLAN trunking

This exercise extends the VLAN configuration from the previous section. The host-facing ports remain **access ports** because end devices such as PCs, printers, and servers typically do not understand VLAN tagging and therefore expect to send and receive **untagged frames** within a single broadcast domain. The uplinks from the access switches to the core and the links between the core switches are configured as **trunk ports** as shown in Figure 3. These ports carry traffic for several VLANs over a single connection, using **tagging** (for example, **IEEE 802.1Q**) to distinguish between them.

Find the IEEE 802.1Q tag format and paste a diagram of the header.

Explain the fields of the tag and what they are used for.

4.1 Configure VLAN trunking

Now, configure the trunk ports as illustrated in Figure 3. Make these ports tagged members of all VLANs and do not change the configuration for the other ports on the access switches. For the core switches, it is important to trunk only the ports indicated in the figure, while keeping the remaining ports untagged in VLAN1 so that management access to the core switches remains easily available.

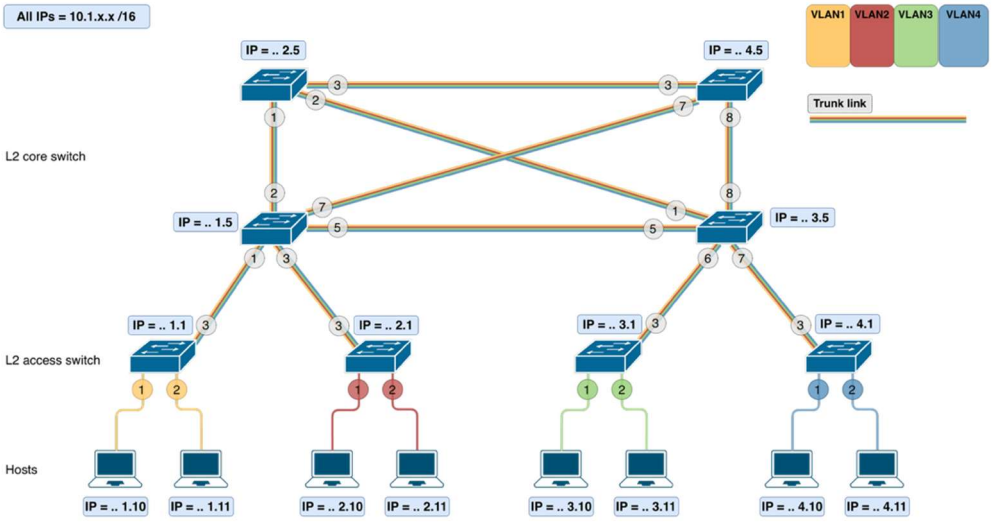


Figure 3: VLAN trunking configuration

Why do uplinks between access and core switches become trunks, and the other ones stay the way they are?

What does tagged mean on a switch port?

What does untagged mean on a switch port?

_____

_____

What VLANs do you need to tag for the switch ports? What happens when you do not tag a certain VLAN?

_____

_____

4.2 Test connectivity

Try to connect. First, ping between hosts in the **same VLAN on the same switch.** Then try hosts that are in the **same VLAN but connected to different access switches**. Finally, test connectivity between hosts that are in **different VLANs**. Observe which connections succeed and which fail.

Which hosts can communicate with each other, and why?

_____

_____

Why is there no connectivity between VLANs? I.e., VLAN 1 to VLAN 2?

_____

_____

4.3 Setting up port mirroring

Next, we will capture packets that travel over a **trunk link**. However, you cannot simply plug your laptop into any switch port and expect to see this traffic. Switches only forward frames to the port where they are needed. To observe traffic from another port (such as a trunk), we need to use **port mirroring**. Port mirroring (also called SPAN) duplicates all incoming and outgoing traffic from one port (the *source*) to another port (the *destination*). You can then connect a laptop running Wireshark to the destination port to analyze the copied traffic [4].

In OpenWrt, there is not always a straightforward GUI option to configure port mirroring on all devices. Therefore, we use the Linux `tc` **(Traffic Control)** tool. The tc tool is normally used for traffic shaping (limiting bandwidth), packet scheduling, and filtering traffic. In this lab, we use tc to **intercept packets on one interface and mirror (copy) them to another interface**. This effectively allows us to implement **software-based port mirroring**, enabling us to inspect traffic that would otherwise not be visible.

Suppose you want to capture packets on your own access switch and specifically want to observe frames that contain an 802.1Q VLAN tag. Which port should you mirror, and to which port should the mirrored traffic be sent?

_____

_____

Before configuring mirroring, you need to know the names of your switch ports by using the following command:

```
ip -br link
```

You should see interfaces such as `lan1`, `lan2`, `lan3`, and so on. In our case, we will mirror from `lan3` to `lan2` as an example.

```
ip link set up dev lan2
tc qdisc del dev lan3 clsact 2>/dev/null
tc qdisc add dev lan3 clsact
tc filter add dev lan3 ingress matchall action mirred egress mirror dev lan2
tc filter add dev lan3 egress  matchall action mirred egress mirror dev lan2
```

What that does:

- `dev lan3` → the **source port** you want to observe
- `mirror dev lan2` → the **destination monitor port** where the copied packets are sent
- `ingress` + `egress` → copy both traffic entering and leaving port 3
- `mirred egress mirror dev lan2` → Linux `tc`'s packet mirror action that transmits the copy out of port 2.

So in plain language: **anything seen on Ethernet port 3 gets copied to Ethernet port 2**, where your Wireshark laptop should be plugged in.

After finishing your capture, you can remove the configuration with the following command:

```
tc qdisc del dev lan3 clsact
```

Capture at least one packet containing an 802.1Q VLAN tag and paste a screenshot of your Wireshark capture below.

_____

_____

Why are VLAN tags not visible on access ports?

_____

_____

What VLAN ID do you observe in your captured packet? Can you relate it to your configuration?

_____

_____

What would happen if you mirrored a non-trunk (access) port instead?

_____

_____

5 Exercise 3: Inter-VLAN routing

At the end of the previous exercise, every VLAN was a perfectly isolated network: hosts in the same VLAN could reach each other across the trunk links, but hosts in different VLANs could not communicate at all. A Layer 2 switch cannot solve this. Each VLAN is a separate broadcast domain and will therefore get its own IP subnet, so traffic between VLANs has to be **routed**. In Lab 2, you connected different subnets by giving the router one physical port per subnet. That approach does not scale to VLANs: we have four VLANs, but the OpenWrt router only has three LAN ports and one WAN port.

Instead, we use a **router on a stick**. The router is connected to the switched network with a **single trunk link** and creates one **logical sub-interface per VLAN** on top of that link. Each sub-interface gets an IP address in the subnet of its VLAN and acts as the **default gateway** for the hosts in that VLAN. A packet between two VLANs travels to the router tagged with the VLAN of the sender, is routed, and travels back over the same cable tagged with the VLAN of the destination.

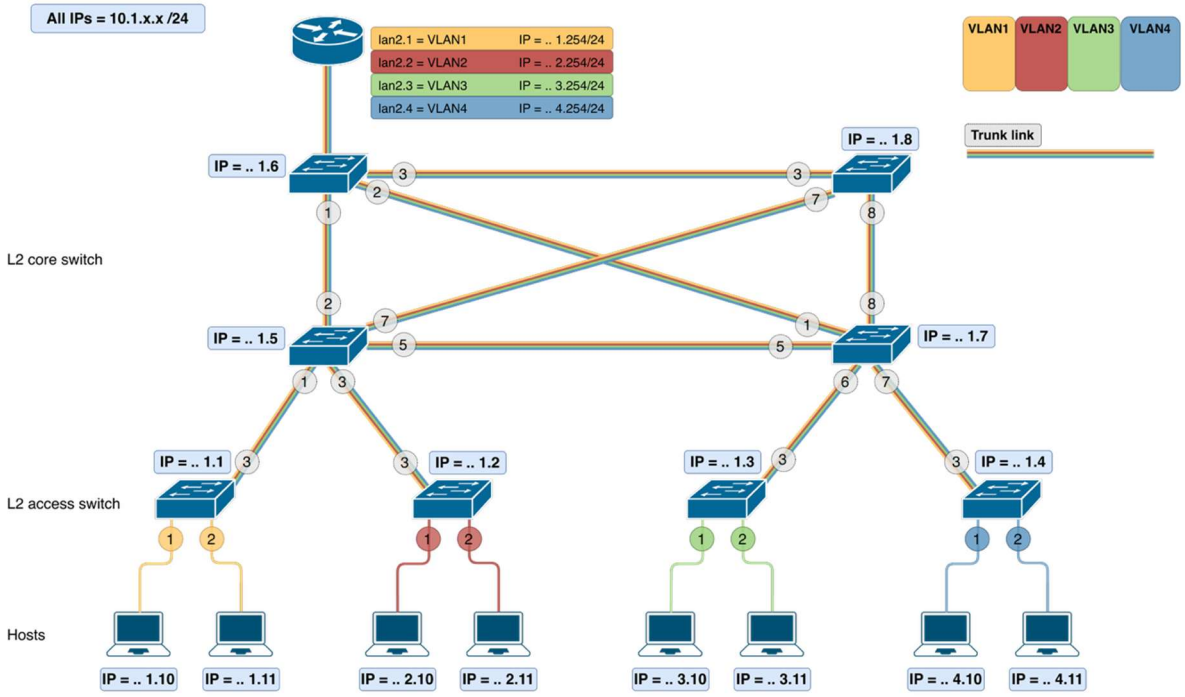


Figure 4: Router on a stick

If a single trunk link already carries traffic for all four VLANs, why do we still need a router at all? Isn't the traffic already there?

5.1 Update the addressing plan

Until now, every host used a `/16` subnet mask (`10.1.x.x/16`). For routing to work, every VLAN must correspond to its own IP subnet, small enough for the router to tell them apart. Conveniently, your hosts are already addressed per VLAN (`10.1.1.x` for VLAN 1, `10.1.2.x` for VLAN 2, and so on), so you only need to shrink the subnet mask to `/24` and add a default gateway, instead of renumbering anything. The router gets address `.254` in every subnet and is the default gateway of that VLAN.

VLAN	Subnet	Hosts	Default gateway
VLAN 1	10.1.1.0/24	10.1.1.10, 10.1.1.11	10.1.1.254
VLAN 2	10.1.2.0/24	10.1.2.10, 10.1.2.11	10.1.2.254
VLAN 3	10.1.3.0/24	10.1.3.10, 10.1.3.11	10.1.3.254
VLAN 4	10.1.4.0/24	10.1.4.10, 10.1.4.11	10.1.4.254

Why did a /16 mask work fine in Exercises 1 and 2, even though hosts in different VLANs technically shared the same /16 network on paper?

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_____

Now that a router is present, what would go wrong if you left the mask at /16 on every host?

_____

_____

Note that the switches are all managed through VLAN1, so their management interface is in the VLAN1 broadcast domain, regardless of the number of the switch. However, the management IP addresses of most switches (for example, `10.1.2.1` or `10.1.3.5`) belong to the subnets of VLAN2, VLAN3, and VLAN4. Change the management IP address of every switch to an address in the VLAN1 subnet, with a `/24` mask, according to the table below. Remember to do this from a VLAN1 port.

Switch	Old address	New address
Access switch 1 to 4	10.1.N.1	10.1.1.N **(10.1.1.1 to 10.1.1.4)**
Core switch 1 to 4	10.1.N.5	10.1.1.(N+4) **(10.1.1.5 to 10.1.1.8)**

Why do the management IP addresses of the switches have to change? What would go wrong if you left them as they are?

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_____

After configuring all hosts with a `/24` mask and the default gateway of their VLAN, check that hosts in the same VLAN can still reach each other, also across different access switches: plug host A of access switch 3 into port 6 of its switch (VLAN1) with the correct subnet address and ping the VLAN1 hosts on access switch 1. Then open Wireshark on your host and ping a host in another VLAN.

What do you see in Wireshark, and why does the ping fail?

_____

_____

5.2 Configure and connect the router

Choose one of your core switches and one of its free ports. Under Bridge VLAN filtering, configure this port as a tagged (T) member of VLAN 1, 2, 3, and 4, exactly like the trunk ports from the previous exercise. Do not give it an untagged/PVID VLAN: every frame the router sends or receives on this link must carry an 802.1Q tag. Connect a cable from this switch port to a LAN port of your OpenWrt router.

Now configure the router. Start by removing LAN2 from the bridge: Go to Network → Interfaces → Devices, configure br-lan, and remove lan2 from the bridge ports.

Next, create one **802.1Q device** per VLAN on top of lan2:

- On the **Devices** tab, click **Add device configuration…**
- Choose **VLAN (802.1Q)** as device type, **lan2** as base device, and enter the VLAN ID. The device name becomes lan2.1, lan2.2, and so on.
- Repeat this for VLAN 1, 2, 3, and 4.

Finally, create one interface per VLAN:

- On the **Interfaces** tab, click **Add new interface…**
- Name it **vlan1** (vlan2, vlan3, vlan4), choose **Static address** as protocol, and select the matching device (lan2.1, lan2.2, …).
- Configure the IPv4 address (10.1.n.254) and netmask (255.255.255.0) according to the addressing plan. Do not configure a gateway and do not enable DHCP.

Why do you remove lan2 from the bridge?

_____

_____

Why do all four sub-interfaces share the same physical port and the same MAC address, yet the router still treats them as four separate networks?

_____

_____

Verify your configuration on the router with the following commands:

```
cat /proc/net/vlan/config
ip addr
ip route
```

What does the output of the first command tell you? How many VLAN devices do you see, and on which physical port are they created?

_____

_____

Which routes do you see in the routing table? Did you have to add static routes or a routing protocol, like in Lab 2? Why (not)?

_____

_____

5.3 Test connectivity

Wait a moment after connecting the router, so that the spanning tree can settle (see Lab 3). Then test connectivity from your hosts using `ping` and `fping`.

Who can you reach, and why?

- The gateway of your own VLAN (`10.1.n.254`)? _____
- The gateway of another VLAN, for example `10.1.3.254` from a host in VLAN2? _____
- A host in the same VLAN, but on another access switch (for example, the moved host A of access switch 3)? _____
- A host in another VLAN? _____
- The management IP of a switch (`10.1.1.y`) from a host in VLAN1? _____
- The management IP of a switch from a host in VLAN2, VLAN3, or VLAN4? _____

Fix the last problem by configuring the default gateway of the switches: **Network → Interfaces → LAN → Edit** and enter `10.1.1.254` as the IPv4 gateway.

Can you now reach the switches from every VLAN?

_____

_____

Why can one physical cable between the switch and router carry traffic for four different VLANs?

_____

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What happens if VLAN3 is not tagged on the switch port connected to the router?

_____

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What happens if the VLAN3 interface exists on the router but has the wrong IP address or subnet mask?

_____

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Use traceroute from a host to a host in the same VLAN and to a host in another VLAN.

How many hops do you see, and which IP addresses show up?

_____

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5.4 Analyze the traffic

Now we will look at what actually happens on the stick. Configure port mirroring on the **core switch** as you did in the previous exercise, with the port connected to the router as source and a free port as destination for your laptop running Wireshark. Choose two hosts in different VLANs, for example, host A of access switch 2 (VLAN2) and host A of access switch 3 (VLAN3). Clear the ARP cache of the host:

```
sudo arp -d -a
```

Start the capture and ping the other host. In Wireshark, you can use the filter vlan.id == 2 to select a specific VLAN.

In which VLAN is the ARP request for the default gateway sent? Would a host in VLAN4 receive this request?

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_____

How many times do you see each ICMP echo request on the link to the router? Explain.

_____

_____

Compare the frame that enters the router with the frame that leaves the router, and complete the table.

Field	Echo request towards the router	Echo request leaving the router
VLAN ID	…	…
Source MAC	…	…
Destination MAC	…	…
Source IP	…	…
Destination IP	…	…
TTL	…	…

References

[1] Practical Networking, "Virtual Local Area Networks (VLANs)," [Online]. Available: https://www.practicalnetworking.net/stand-alone/vlans/.

[2] Practical Networking, "Routing Between VLANs & Layer 3 Switches," [Online]. Available: https://www.practicalnetworking.net/stand-alone/routing-between-vlans/.

[3] Cisco, "Inter-Switch Link and IEEE 802.1Q Frame Format," 25 Augus 2006. [Online]. Available: https://www.cisco.com/c/en/us/support/docs/lan-switching/8021q/17056-741-4.html.

[4] "Switches - Port mirroring," [Online]. Available: https://svanheule.net/switches/testing/mirroring.